

Professional issue

Artificial intelligence and machine learning | applications in musculoskeletal physiotherapy

Christopher Tack

Guy's and St Thomas' NHS Foundation Trust, Guy's Hospital, Great Maze Pond, SE1 9RT, London, UK

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ABSTRACT

Introduction: Artificial intelligence (AI) is a field of mathematical engineering which has potential to enhance healthcare through new care delivery strategies, informed decision making and facilitation of patient engagement. Machine learning (ML) is a form of narrow artificial intelligence which can be used to automate decision making and make predictions based upon patient data.

Purpose: This review outlines key applications of supervised and unsupervised machine learning in musculoskeletal medicine; such as diagnostic imaging, patient measurement data, and clinical decision support. The current literature base is examined to identify areas where ML performs equal to or more accurately than human levels.

Implications: Potential is apparent for intelligent machines to enhance various areas of physiotherapy practice through automatization of tasks which involve data analysis, classification and prediction. Changes to service provision through applications of ML, should encourage physiotherapists to increase their awareness of and experiences with emerging technologies. Data literacy should be a component of professional development plans to assist physiotherapists in the application of ML and the preparation of information technology systems to use these techniques.

1. Introduction

Artificial intelligence (AI) can enhance healthcare through advancement of care delivery, decision-making, and patient engagement (House of Lords, 2018). AI is the development of computer systems to solve problems associated with human intelligence (Fogel and Kvedar, 2018). “Narrow AI” (nAI)- intelligence in a specific area-is the currently used application of AI in society; and differs from theoretical “general AI” which simulates human-level intelligence across multiple domains (Franklin, 2007).

Machine learning (ML) is a form of nAI (Peek et al., 2015) where algorithms make predictions to interpret data and “learn”, without static program instructions (Ayodele, 2010). There are two main forms of ML: supervised and unsupervised. Supervised ML is where algorithms are given training data, which is analysed for features important for classification and labelled. The model is then “trained” with this data before being tested with unlabelled data. For example, if the data are X-ray images, they are first labelled by a radiographer so the model can learn how to interpret. The model is then given unlabelled data and tested to give an output of interpretation. When the output is discrete (e.g. fracture present on X-ray vs no fracture) this is called “Classification” (prediction of a label). If the output is continuous (e.g. range of

motion measurement) this is “Regression” (prediction of a quantity). Unsupervised ML is used to identify patterns without training. Common forms are cluster analysis (where data is grouped by patterns of characteristics) or association (where rules are discovered by which data is governed). Healthcare is suited for nAI, as complex datasets from electronic health records (EHR) provide challenging classification opportunities (Krittanawong, 2018). Benefits through pathway refinement and task automation can also reduce costs (Harwich and Laycock, 2018). This issue aims to outline current ML applications in the musculoskeletal field.

2. Supervised learning

2.1. Medical imaging

Fig. 1a illustrates supervised learning classification. It is essential that the ML model has sufficient labelled data from which to learn. Various forms of data are used in healthcare, and the abundance of diagnostic images collated makes them a prime target for classification (Berg, 2017). Table 1 shows the accuracy of supervised learning models across various areas of musculoskeletal medicine. For example, supervised learning can be used to teach ML models how to classify X-ray

E-mail addresses: Christopher.tack@gmail.com, Christopher.tack@gstt.nhs.uk, @allpowersphysio.

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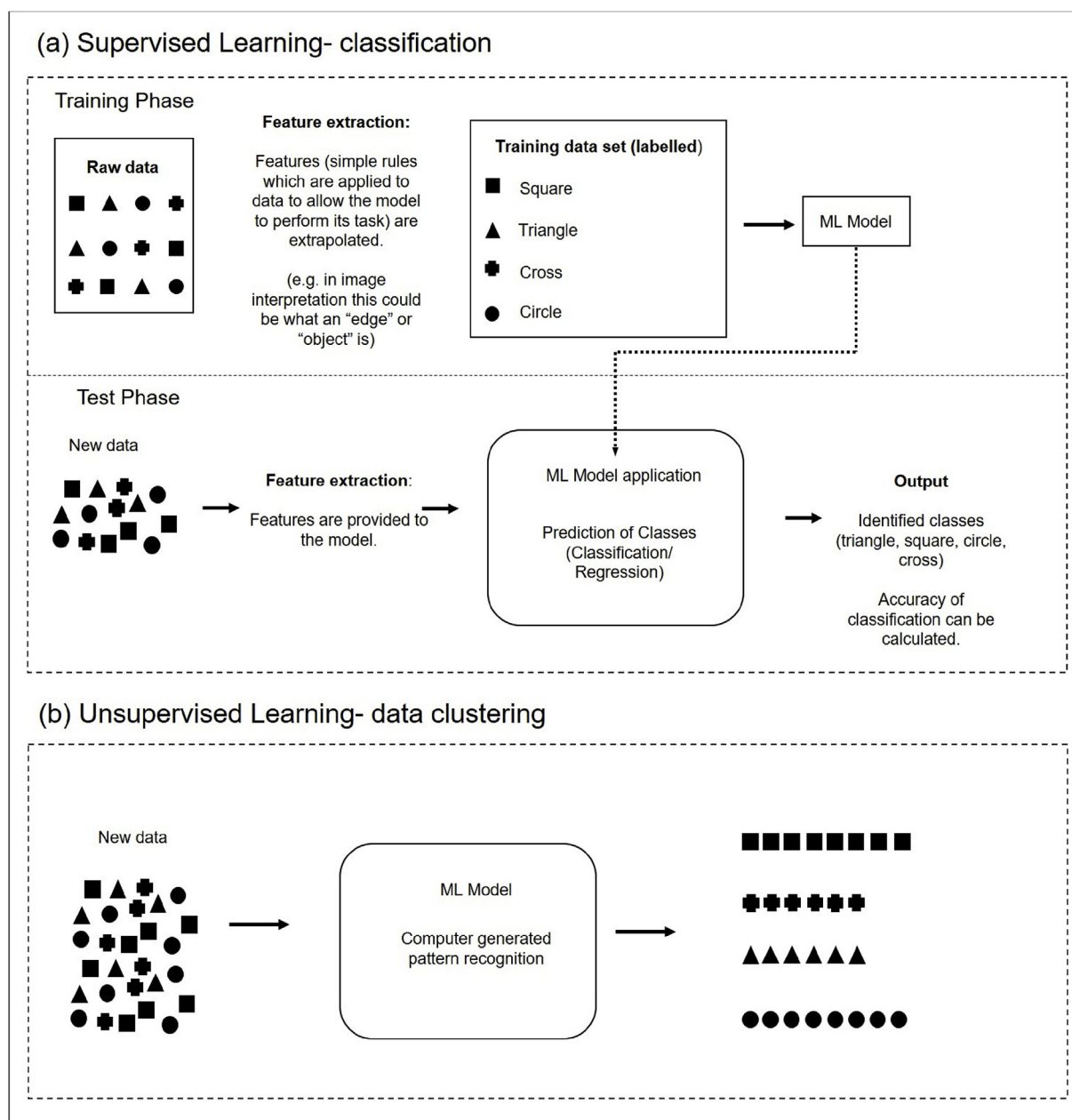


Fig. 1. Machine learning data processing examples (a) Supervised and (b) unsupervised learning.

and MRI images with accuracy equal to or greater than human experts (Jamaludin et al., 2017; Olczak et al., 2017); including whether pathophysiological changes (e.g. fracture or spinal degeneration) are present (Fig. 2). Similarly, models can be taught how to evaluate prospective MRI images of knee cartilage (Ashinsky et al., 2017), and predict the onset of arthritis 36 months before it is identifiable to human observers (75% accuracy). However, whilst automated image interpretation can be accurate, predictive and tireless; it does require large training datasets. This is particularly true when using complex neural network algorithms (systems based on mathematical models of the brain) like the above studies (Razzak et al. 2018). For example, Olczak et al. (2017) used 256,000 x-ray images to train their neural network to classify images. If annotated, anonymised image datasets could be made open-access, then superior automated diagnostic accuracy could attenuate service inequality; standardise image interpretation across healthcare institutions (Harwich and Laycock, 2018), and be an asset in settings without radiological expertise (Berg, 2017).

2.2. Pain

With long-standing pain prevalence of 30–50% in the UK (Fayaz et al., 2016), another application of supervised ML is in the classification of individuals to pain phenotypes based on brain MRI. Pain present without correlative tissue pathology, and reliance on self-reporting for subgrouping, makes identification of the neural correlates of pain an important challenge (Boissoneault et al., 2017). Lötsch and Ultsch (2018) summarise ML applications used in longstanding pain (not exclusively musculoskeletal); outlining the use of ML to classify subjects to a predicted pain phenotype. The review examines experimental pain (Chesler et al., 2002); fibromyalgia (Sevel et al., 2016); and low back pain (LBP) (Jiang et al., 2017); but provides no conclusions regarding effectiveness of ML classification.

ML methods show variable accuracy in discriminating between those with a pain condition and healthy individuals, based upon brain MRI (Boissoneault et al., 2017). Biomarkers (e.g. grey matter volume/density or functional connectivity between regions) can be used for

Table 1
Accuracy of supervised machine learning techniques for musculoskeletal applications.

	Classification question	Data source	Classification label	Classification Accuracy [%]	Algorithm used
Olczak et al. (2017) Jamaludin et al. (2017)	Is pathology present or not?	X-ray	Fracture (vs no fracture)	83	16 layer CNN CNN
		MRI	Pfirmann grade	70.4	
			Disc narrowing	75.4	
			Spondylolisthesis	95.4	
			Central canal stenosis	94.7	
Ashinsky et al. (2017) Bagarinao et al. (2014) Ung et al. (2012) Robinson et al. (2015)	Is pain neuromapping phenotype identifiable with clinical diagnosis?	MRI	Cartilage mapping	75	WND-CHRM SVM SVM LR, MP, Bayes, SVM, j48 DT
		Structural MRI	CPP	73	
			CLBP	76	
			FM	76	
		fMRI	CLBP	92	
Callan et al. (2014) López-Solà et al. (2017) Burns et al. (2018)	Can successful exercise performance be identified?	Inertial sensor	FM	93	SLR SVM, LR CNN k-NN SVM
			Accurate exercise performance	99.4	
		Inertial sensor		97.8	
				94.1	
Kianifar et al. (2017)	Can risk of injury be classified based upon movement quality?	Inertial sensor	Risk of injury with movement:	95	10F-CV 10F-CV LOSO-CV LOSO-CV
			“high” vs “low”	72	
			“high” vs “moderate” “low”	90	
			Accuracy of performance:	60	
			“poor” vs “good”		
Nijeweme-d'Hollosy et al. (2018)	Can CLBP subgroups be stratified accurately?	EHR	“poor” vs “moderate” vs “good”		DT BT
			“Physiotherapy” vs “GP” vs “self-management”	71.05	
				71.05	

10F-CV- 10-fold cross-validation.

BT-boosted tree.

CLBP- chronic low back pain.

CNN- convolutional neural network.

CPP- chronic pelvic pain.

DT-decision tree.

EHR-electronic health records.

FM-fibromyalgia.

K-NN- k-nearest neighbour.

LOSO-CV- leave-one-subject-out cross-validation.

LR-logistical regression.

MP- multilayer perceptron.

SVM-support vector machine.

WND-CHRM- Weighted Neighbour Distances using a Compound Hierarchy of Algorithms Representing Morphology.

classification; acting as a surrogate to self-reporting (Mackey, 2013). With pain being heterogeneous, and a complex interplay of etiological, genetic and environmental factors (von Hehn et al., 2012); the implications of these studies are debatable. Furthermore, functional MRI shows greater accuracy but currently has limited use in the routine assessment of pain (Cowen et al., 2015). Table 1 outlines the accuracy of ML to identify brain phenotypes patterns for various pain conditions (e.g. LBP, fibromyalgia). Nevertheless, the potential for clinical application of ML-classified MRI for anatomical phenotyping of pain may develop in the future (Borsook and Becerra, 2007).

2.3. Wearable technology

Wearable technology offers a source of rich, epidemiological data through surveillance of physical behaviour; assistive in managing chronic disease (Phillips et al., 2018). Exercise adherence is often poor, despite being predictive of successful rehabilitation (Holden et al., 2014). Data collected via inertial sensors in wearable technology can be classified to identify whether users are accurately performing and adherent to exercise. Burns et al. (2018) tested performance accuracy in an asymptomatic population (n = 20) who performed a rotator-cuff exercise protocol (Kuhn et al., 2013) whilst wearing an Apple iWatch. Various methods of supervised learning were used to classify the accuracy of exercise (compared to an established dataset of “successful” performance); showing excellent accuracy across all algorithms. 99.4% classification accuracy was seen with a neural network, indicating

promise for the use of wearable technologies and ML for exercise monitoring. As exercise performance has variable barriers to adherence (Jack et al., 2010), it is likely that measurement of performance through wearable technology alone, will be insufficient to optimise adherence. However, potential exists to use such data to identify and modify patient behaviour for clinical benefit.

2.4. Risk prediction

Frontal plane knee biomechanics are associated with knee injury risk prediction (Culvenor et al., 2014). Kianifar et al. (2017) used inertial sensor data to classify single-leg squat performance according to degree of knee valgus (n = 14, 140 images) and 3 expert raters' opinions of potential risk. They used supervised learning to classify between 3-classes (“poor”, “moderate” and “good”). The results found that accuracy was high during 2-class classification, but reduced by 30% with added complexity of 3-class classification. Supervised learning regression has also been applied to predict internal knee abduction moment based upon lower limb joint angles on video images during gait (Aljaaf et al., 2016). 3131 images were analysed, and various ensemble (multi-layer) learning strategies demonstrated good accuracy of estimating abduction moment. A neural network method, despite long training time, was deemed the best predictor of knee force. These studies show that visual and inertial sensor data can be analysed by ML to predict injury risk patterns associated with dynamic knee valgus. However, with injury risk being multifactorial, biomechanical

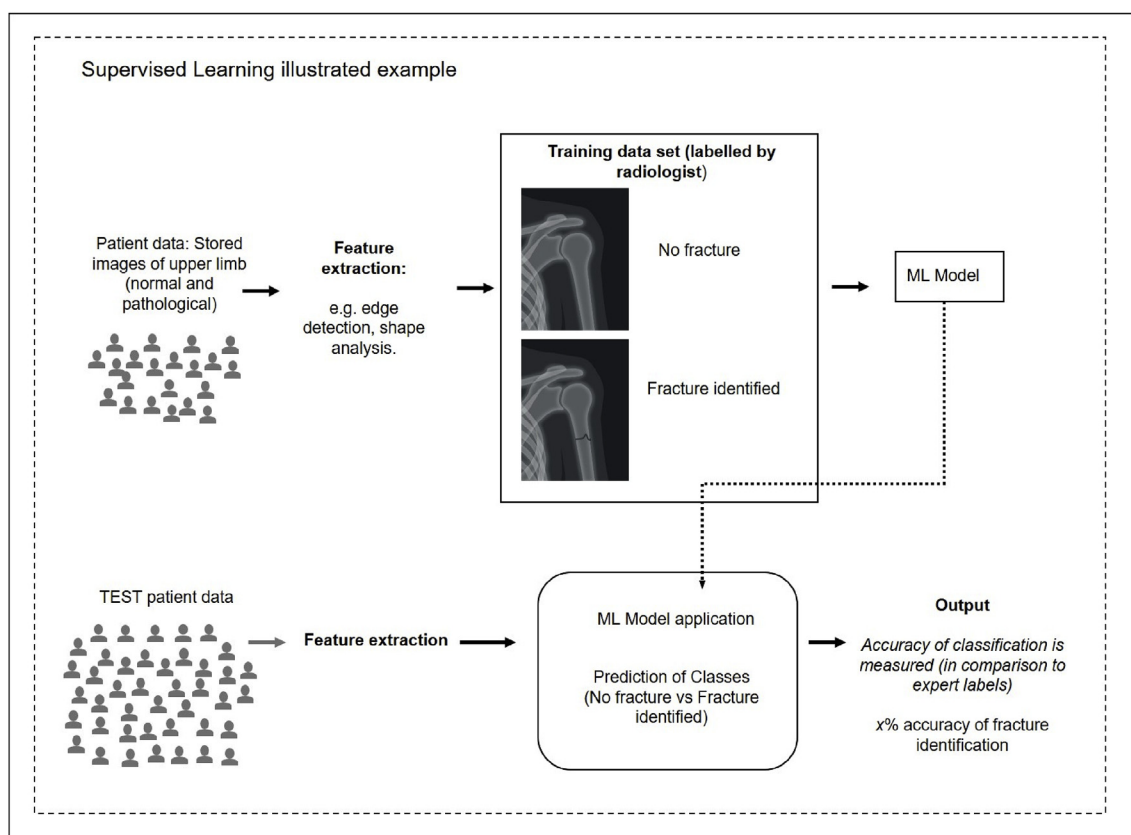


Fig. 2. Supervised learning illustrated example with fracture identification on x-ray.

data is likely only one part of injury prevention management (Murphy et al., 2003).

2.5. Decision support

Clinical decision support systems (CDSS) provide recommendations on diagnosis and treatment (Musen et al., 2014). Systems have been developed for LBP: for example the StartBack risk stratification tool which identifies prognostic indicators to classify individuals into risk groups (Hill et al., 2008). Nijeweme-d'Hollosy et al. (2016) developed a digital CDSS to stratify patients to self-management, GP attendance or self-referral to physiotherapy. An ontology and decision tree to classify subjects was developed according to 43 decision factors; such as general factors (e.g. occupation), 'psychosomatic' factors (e.g. depression, kinesiophobia); and serious pathology signs (i.e. red flags). Subsequently, supervised ML algorithms were applied to classify individuals using the CDSS (Nijeweme-d'Hollosy et al. 2018); finding all classified better than random allocation. This demonstrates the CDSS can successfully be used with ML to classify subjects, and further progress could make the combination more rigorous than by human decision-making. Comparably, human ability to delineate patients into low vs medium/high risk via StartBack is between 72.1 and 78.1% sensitivity and 42.91–75% specificity (Pagé et al., 2015). Thus, ML may provide more accurate allocation to services in the future, whilst increasing accessibility and speed of self-referral.

3. Unsupervised learning & data mining

There are few examples of unsupervised learning methods being used in the musculoskeletal field. The Chronic Pain Challenge (Navani and Li, 2016) predicts the risk of pain chronicity according to weighted values for health behaviours. It uses supervised and unsupervised methods, showing accurate prediction of pain (visual analogue scale)

and Oswestry Disability Index; according to associated scores for depression, nutrition and physical activity. However, whilst this highlights the potential for ML to classify risk of chronicity based upon patient-reported data, the accuracy of unsupervised learning alone is not established.

Digitisation makes healthcare data accessible by computers (Bell, 2017; The Royal Society, 2017), with ML reliant on large datasets to accurately predict (Peek et al., 2015). Unsupervised learning is often applied within data-mining (the examination of large databases) in order to make new discoveries regarding risk or causation, where patterns are not obvious to clinicians (Obenshain, 2004). Such means have been used to identify symptoms predictive of chronic fatigue syndrome from diverse data sources (Watson et al., 2014). Ensuring data format, quality and security is essential to the adoption of ML, to prevent bias being placed into algorithms, and maintaining public trust in the use of health data (Hart, 2017; Harwich and Laycock, 2018; House of Lords, 2018). Health surveillance systems raise concerns over data privacy, however trends can be detected even with anonymised data (Obenshain, 2004). Physiotherapists should engage with digital transformation by becoming more data-literate, and professional development programmes should include basics of data science to prepare clinicians for future practice (House of Lords, 2018). Physiotherapists should become involved in the standardisation of diagnostic coding procedures to aid data usability (Williams et al., 2017); as well as assisting development of bespoke IT platforms, rather than reactive, plug-in alterations often seen in healthcare (Harwich and Laycock, 2018; Watson, 2016). This should facilitate interoperability within (and between) institutions to aid data sharing which is essential for unsupervised learning methods (Peek et al., 2015).

4. Conclusions

Supervised ML has potential to evolve physiotherapy practice,

through super-human level diagnostics, decision making and measurement. For this, data sharing is paramount to train systems effectively. Unsupervised ML is unproven but may assist data-mining to identify patterns of association within patient populations. Physiotherapists should engage with technological innovation in practice, to guide its clinical application.

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